Quiz-2

1| 3 DOF → 3 degrees of freedom → 3 active joints

5 DOF → 5 degrees of freedom → 5 active joints

2| based on mobility → 2 types

- active
- passive

based on movement → 2 types

- prismatic
- revolute

3| • DH- Parameters

i	a	d	α	θ

1	0	5	90°	35°
2	4	0	0	50°
3	8	0	0	0°
4	7	0	0	-140

4)

For joint 1.

$$T_1 = \begin{bmatrix} \cos 35 & -\cos 90 \sin 35 & \sin 90 \sin 35 & 0 \cos 35 \\ \sin 35 & \cos 90 \cos 35 & -\sin 90 \cos 35 & 0 \sin 35 \\ 0 & \sin 90 & \cos 90 & 5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_1 = \begin{bmatrix} 0.82 & 0 & 0.57 & 0 \\ 0.57 & 0 & -0.82 & 0 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

For joint 2.

$$T_2 = \begin{bmatrix} \cos 50 & -\cos 0 \sin 50 & \sin 0 \sin 50 & 4\cos 50 \\ \sin 50 & \cos 0 \cos 50 & -\sin 0 \cos 50 & 4\sin 50 \\ 0 & \sin 0 & \cos 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_2 = \begin{bmatrix} 0.64 & -0.77 & 0 & 2.57 \\ 0.77 & 0.64 & 0 & 3.06 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

For joint 3.

$$T_3 = \begin{bmatrix} \cos 0 & -\cos 0 \sin 0 & \sin 0 \sin 0 & 8\cos 0 \\ \sin 0 & \cos 0 \cos 0 & -\sin 0 \cos 0 & 8\sin 0 \\ 0 & \sin 0 & \cos 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_3 = \begin{bmatrix} 1 & 0 & 0 & 8 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

For joint 4.

$$T_4 = \begin{bmatrix} \cos(-140) & -\cos 0 \sin(-140) & \sin 0 \sin(-140) & \cos(-140) \\ \sin(-140) & \cos 0 \cos(-140) & -\sin 0 \cos(-140) & \sin(-140) \\ 0 & \sin 0 & \cos 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_4 = \begin{bmatrix} -0.77 & 0.64 & 0 & -5.36 \\ -0.64 & -0.77 & 0 & -4.5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T = T_1 \times T_2 \times T_3 \times T_4$$

$$= \begin{bmatrix} 0 & 0.82 & 0.57 & 6.3 \\ 0 & 0.57 & -0.82 & 4.4 \\ -1 & 0 & 0 & 7.2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\therefore (x, y, z) = (6.3, 4.4, 7.2)$$

- Trigonometry

Side view

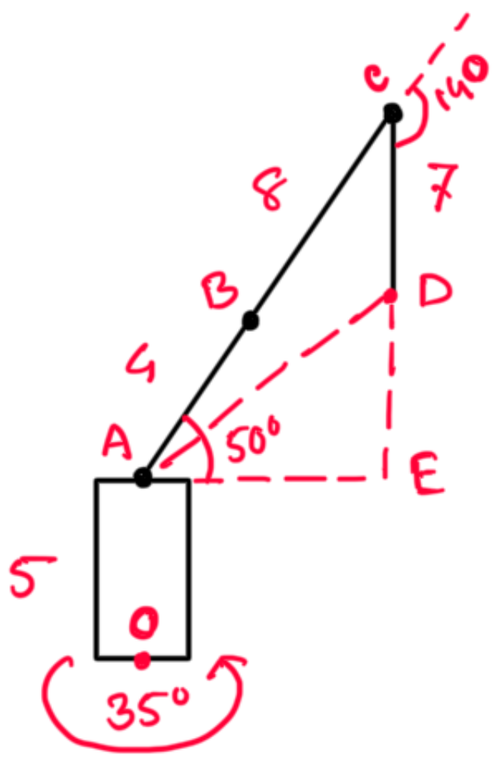
$\triangle ACD$,

$$\angle ACD = 180 - 140 = 40^0$$

$$\therefore AD^2 = AC^2 + CD^2 - 2 \cdot AC \cdot CD \cdot \cos \angle ACD$$

$$\Rightarrow AD = \sqrt{12^2 + 7^2 - 2 \cdot 12 \cdot 7 \cdot \cos 40}$$

$$= 8.02$$



And,

$$\angle CAD = \cos^{-1}\left(\frac{AC^2 + AD^2 - CD^2}{2 \cdot AC \cdot AD}\right)$$

$$= \cos^{-1}\left(\frac{12^2 + 8.02^2 - 7^2}{2 \times 12 \times 8.02}\right)$$

$$= 34.13^0$$

$$\therefore \angle DAE = 50 - 34.13 = 15.87$$

ΔDAE,

$$\cos \angle DAE = \frac{AE}{AD}$$

$$\Rightarrow AE = \cos(15.87) \times 8.02 = 7.71 \quad \Rightarrow \text{projection}$$

and,

$$\sin \angle DAE = \frac{DE}{AD}$$

$$\Rightarrow DE = \sin(15.87) \times 8.02 = 2.19$$

$$\therefore Z = 5 + 2.19 = 7.19$$

Top View

$$\cos\angle EOF = \frac{OF}{OE}$$

$$\Rightarrow OF = \cos(35) \times 7.71$$

$$= 6.3 \Rightarrow x$$

$$\sin\angle EOF = \frac{EF}{OE}$$

$$\Rightarrow EF = \sin(35) \times 7.71$$

$$= 4.4 = y$$

$$\therefore (x, y, z) = (6.3, 4.4, 7.19)$$

